

Course Learning Objectives
CLO1: Share a reproducible R code workflow using the desktop RStudio. Share an R documentation that can be re-run on a different machine, with dated files, clear naming, documented steps/assumptions, metadata/citation for data, and other resources.
CLO2: Confidently problem-solve R coding tasks. Operate RStudio (install/load/update packages), write and run scripts without error, keep evidence of bugs, narrate the troubleshooting journey, and compare at least one alternative approach (e.g., arguments, indexing styles, base vs. tidy verbs later).
CLO3: Demonstrate learning skills & AI competencies in workflow. Demonstrate growth in two learning virtue clusters via memos: • Inquiry & Agency (Curiosity + Courage + Autonomy) • Resilient Rigor (Tenacity + Thoroughness + Open-mindedness) Use AI appropriately as a learning aid (explanations, error sense-making) with brief disclosure; type all code yourself.
CLO4: Next R Course Readiness via Signature Assignment. Produce an end-to-end mini-analysis on a chosen or provided dataset: raw data → wrangled RDS → knit HTML/PDF (Rmd/Qmd) with memos, data license/source, and a short oral/screencast walkthrough.

Module 1	
Title	R: Navigation; LS: Why R & Tenacity; AI: AI Course Policy
CLOs, MLOs	R: • Core ◦ Set up with R and RStudio on computer ◦ Able to Navigate R Studio ◦ Able to create, save, and submit an R script file to Canvas ◦ Manage R Environment ◦ Store objects • Advanced ◦ Custom settings in R ◦ Produce an RHistory report LS: • Core ◦ Write memos in R script without run errors ◦ Problem-Solving Errors ◦ Functions and Help tab • Advanced ◦ Communitate why learn R ◦ Tenacity Learning Skill AI: • Core

	o AI Course Policy
Reading / Video / Content	Part 1 1. Syllabus / AI Course Policy - a. 2. Set up R and R Studio / Navigate R Studio a. Reading: Data Analysis and Visualization in R for Ecologists i. 1. Introduction to R and RStudio (<- entire 1 page) b. Reading: DataQuest c. Reading: Irizarry: Introduction d. Course Lesson "Install R" e. Course Video: How to install R & R Studio (Windows) f. YouTube: @RProgramming101: How to install R & R Studio Video (Mac) 3. R Scripts a. Course Lesson: R Scripts, memoing and help ("Lesson 1" split into two) b. Reading: FIND/WRITE Tenacity i. Learning Skill Blueprint (Kind of old) ii. Ethics in Research c. Course Video: Writing memos, Problem-solving, Help (Tenacity applied) d. Reading: FIND/WRITE memoing for social science research &/or annotating code e. Course Video: Create, save, and submit an R script to Canvas 4. Manage R Environment a. Reading: Irizarry: Saving your Workspace (2.2.6) b. Reading: Managing your environment c. Course Video: Create, save, and submit an RData file to Canvas 5. Why R a. Course Video: Interview Next Course Instructors b. Course Video: Interview Industry Users c. Course Video: Showcase R Capabilities d. YouTube: @RProgramming101: Why you should use R e. YouTube: @RProgramming101: R vs Python f. YouTube: @IBMTechology: R vs Python Part 2 6. Store Objects a. Course Lesson: R Scripts, memoing and help (Need to split "Lesson 1") b. Course Video: Writing memos, Problem-solving, Help (Part 2) c. Reading: R for Social Scientists: Creating objects in R d. Reading: Irizarry: The very basics (2.2.1, 2.2.2, 2.2.3, 2.2.4, 2.2.5) e. Reading: Irizarry: Why use scripts & Commenting your code (2.2.7, 2.2.8) 7. Advanced Content a. Course Lesson: Create: RHistory report b. Course Video: Custom settings in R
Activity or Assessment <i>Part 1</i>	1. Core: NONE: Syllabus / AI Course Policy 2. Core: Quiz: Set up R and R Studio / Navigate R Studio 3. Core: Activity: R Scripts a. Submit R Script with Memos from lesson/video notes 4. Core: Activity: Manage R Environment a. Load / Run Pre-made R Script (Submit RData file) 5. Advanced: Discussion: Why R
Activity or Assessment <i>Part 2</i>	1. Core: Assignment: It's at the end of R Scripts, memoing and help 2. Advanced: Discussion: Why R 3. Advanced: Activity: Advanced Content

	a. Upload a screenshot of the custom setting/RHistory?
Signature Assignment	Introduction to the course project and plan
Module 2	
Title	R: Function Basics; LS: Transparency; AI: Techniques and applications
CLOs, MLOs	R: • CORE ○ Build numeric, integer, character, and logical vectors ○ Check structure, length, and data type ○ Math on numeric vectors ○ Logical operators • Advanced ○ Data Coresion ○ Create dataframes from vectors ○ Index Vectors ○ Create lists LS: • Core ○ Replicable work in an R Notebook with annotations • Advanced ○ Transparency Skill Highlight ○ R Notebook HTML AI: • Core ○ Documenting AI use • Advanced ○ AI techniques and applications: Help resources ○ Prompting Templates
Reading / Video / Content	1. Vector Basics Part 1 a. Course Lesson: "Lesson 2" (Split into two for part 1 and part 2) b. Course Video: c() for chr, num, int c. Reading: Irizarry: Vectors (2.4.1, 2.4.2, 2.4.3) d. Course Video: Length and Class e. Course Video: Math on Vectors f. Reading: Irizarry: Vector Arithmetics (2.8.1, 2.8.2, 2.8.3) g. Reading: R for Social Scientists: Introduction to R i. Creating objects in R (stop at functions and their arguments) ii. Vectors and Data Types 2. Function Basics Part 1Function Basics Part 2 a. Course Lesson: "Lesson 2" (Split into two for part 1 and part 2) b. Course Video: Logical Vectors & Logical Operators c. Course Video: Dataframes from Vectors (Advanced) d. Course Video: Indexing Vectors(Advanced) e. Reading: Irizarry: Subsetting (2.4.4) f. Reading: R for Social Scientists i. Getting Started with R g. Reading: R for Social Scientists: Introduction to R i. Subsetting Vectors (Stop at Missing Data) h. Course Video: Lists (Advanced) i. Reading: Data Analysis and Visualization with R for Genomics

	i. R Basics j. Reading: Data Science A First Introduction i. 3.3 Data frames, vectors, and lists k. YouTube: @RProgramming101: Using functions and objects in R 3. Replicable Work a. Course Video: Formatting an R Notebook (date, code chunks) b. Course Reading: Transparency c. Course Video: Functions and Help Tab d. Reading: Data Analysis and Visualization with R for Genomics i. Getting help with R e. Course Video: Ethics of AI: Transparency of use and guidelines HELP f. Course Video: Write memos in R Notebook g. Course Video: Knit Notebook to HTML
Activity or Assessment <i>Part 1</i>	1. Core: <u>Assignment</u>: Function Basics (Currently in Lesson 2) a. c(), length & class, math on vectors 2. Advanced: <u>Group Activity</u>: Vector Playlab (MadLibs)
Activity or Assessment <i>Part 2</i>	3. Core: <u>Assignment</u>: Function Basics (Currently in Lesson 2) a. Logical vectors, logical operators (Core) 4. Advanced: <u>Group Activity</u>: Vector Playlab (Second Half)
Signature Assignment	None this week
Module 3	
Title	R: Function Foundations; LS: Courage; AI: Ethics of AI
CLOs, MLOs	R: • Core ○ Distinguish between required vs. default arguments ○ Nest functions ○ Detect missing values in vectors ○ Manage computer's folders ○ Import datasets • Advanced ○ Sort and round descriptive statistics ○ Find/download data from open-source resources (Part 1) ○ Securely manage data LS: • Core ○ Review course content • Advanced ○ Courage Skill Highlight ○ Citing resources AI: • Core ○ Identify the privacy risks of certain AI systems • Advanced: ○ Utilize AI within rules and regulations
Reading / Video / Content	Part 1 1. Function Foundations

	a. Course Lesson: "Lesson 3" & "Lesson 4" b. Course Video: Required vs. Default arguments c. Course Video: Nesting Functions d. Course Video: Detecting Missing Values e. Reading: R for Social Scientists: Introduction to R i. Functions and their Arguments f. Course Video: Creation & Basic Manipulation (Advanced) g. Course Video: Descriptive Statistics and Summaries (Advanced) h. Course Video: Sorting, Rounding, formatting (Advanced) 2. Reading: FIND/WRITE Courage Learning Skill / Courage Lab Inspired Part 2 3. Import Data a. Course Lesson: "Lesson 5" b. Course Video: Load Package datasets c. Course Video: Folder Management d. Course Video: Importing CSV e. Course Video: Importing Excel f. Course Video: Save RDS (Advanced) h. Reading: Data Science A First Introduction i. Chapter 2 Reading in data locally and from the web i. Reading: Data Analysis and Visualization with R for Genomics i. R Basics continued - factors and data frames 1. Importing tabular data into R j. Reading: Import Data Cheatsheet k. YouTube: @RProgramming101: Importing data from Excel into RStudio 4. Data Control a. Course Video: Data Resources (Advanced) b. Reading: Boise State Guide to Locating Datasets c. Course Video: Citing Resources (Advanced) d. Course Video: Data Management (Advanced) e. Reading: Best Practices for Research Data Management and Creating a Data Management Plan (Advanced)
Activity or Assessment <i>Part 1</i>	1. Core: <u>Assignment</u>: Function Foundations: in "Lesson 3" & "Lesson 4" 2. Advanced: <u>Activity / Discussion Board</u>: Courage 3. Advanced: <u>Quiz</u>: Part 1 Content (Advanced)
Activity or Assessment <i>Part 2</i>	1. Core: <u>Assignment</u>: Import Data: in "Lesson 5" 2. Advanced: <u>Activity / Discussion Board</u>: Courage 3. Advanced: <u>Quiz</u>: Part 2 Content 4. Advanced: <u>Activity</u>: Submit RDS In "Reproducible Workflow"
Signature Assignment	1. <u>Assignment</u>: Submit raw data with citation (Advanced) -> Due in Module 4 a. Find/download data from open-source resources (Advanced) • Data Is Plural — Structured Archive (Excel) • Library Datasets • Digital Commons Network • https://data.gov/ • D-Place
Module 4	
Title	R: Base R Dataset Manipulation; LS: Curiosity; AI: Human-Centered Mindset

CLOs, MLOs	R - Core - Index datasets - Subset and extract data from datasets - Data wrangle using basic Base R commands - Advanced - Use R Markdown - Find/download data from open-source resources (Part 2) LS - Core - Document thinking process - Advanced - Curiosity Skill Highlight AI - Advanced - Explain the difference between humans controlling AI versus AI controlling humans
Reading / Video / Content	Part 1 - Base R Data Set Exploration - Course Lesson: "Lesson 6" - Course Video: Dataset Indexing: Bracket Syntax - Course Video: Subsetting - Course Video: Extraction - Reading: R for Social Scientists (3. Starting with Data) - Replicable Work - Course Video: Formatting an R Markdown (date, code chunks) - Course Reading: FIND/WRITE Curiosity skill highlight - Vector Lab Inspired - Review Module 1:3 - Reading: R Markdown Cheatsheet - Reading: Bookdown R Markdown - Course Video: Write memos in R Markdown - Course Video: Knit R Markdown to HTML - AI - Course Video: Human-centered Mindset - Course Reading: FIND/WRITE: Human-centered Mindset Part 2 - Base R Dataset Manipulation - Course Lesson: "Lesson 6" - Course Video: Insertion - Course Video: Renaming Columns - Replicable Work - Advanced: Course Reading: FIND/WRITE: Citing Data
Activity or Assessment <i>Part 1</i>	- Core: <u>Assignment</u>: Explorations: in "Lesson 6" - Advanced: <u>Activity / Discussion Board</u>: Vector Lab Inspired Review Module 1:3 Curiosity. - Advanced: Reflection Activity: Values & "Why I'm Learning R" quiz in sandbox course. (combining human-centered mindset and Curiosity)
Activity or Assessment <i>Part 2</i>	Core: <u>Assignment</u>: Manipulations in "Lesson 6"

	2. Cont ... Advanced: Reflection Activity: Values & "Why I'm Learning R" quiz in sandbox course. (combining human-centered mindset and Curiosity)
Signature Assignment	1. Assignment: Submit raw data with citation (Advanced) -> Started in Module 3 a. Data declaration A-B Grade Students find their own dataset, whereas C students use a provided dataset OR create a personal help file. 2. Core: Fire Assignment 3. Personal Help File
Module 5	
Title	R: Logical for Dataset; LS: Autonomy; AI: Human-Centered Mindset
CLOs, MLOs	R - Core - Use \$ to access columns in datasets - Conduct logical tests on columns in datasets - Apply which(), &, , and %in% with dataset wrangling. - Use piping (%>%) with dplyr verbs (select and filter) - Advanced - Build data with sample() and ifelse() - Check rows for duplicates - Randomly sample rows - Use Quarto LS - Core - Formating Documents - Advanced - Autonomy Skill Highlight - Review course content AI - Explain the difference between humans controlling AI versus AI controlling humans
Reading / Video / Content	Part 1 - Logicals - Course Lesson: "Lesson 7" - Course Video: \$ - Course Video: Logical Tests - Course Video: which() - Course Video: & and %in% - Reading: R for Social Scientists (3. Starting with Data) - Replicable Work - Course Reading: FIND/WRITE Autonomy skill highlight - Course Video: Human-centered Mindset - Course Reading: FIND/WRITE Human-centered Mindset Part 2 - dplyr Intro - Course Lesson: "Lesson 8" - Course Video: Pipe %>% - Course Video: select() - Course Video: filter() - Course Video: select() and filter() - Reading: Data Science A First Introduction 1.7 Creatng subsets of dataframes with filter and select

	ii. 3.5 Using select to extract a range of columns iii. 3.6 Using filter to extract rows g. Reading: Data Analysis and Visualization in R for Ecologists i. Manipulating data ii. The Pipe h. Reading: Data Analysis and Visualization with R for Genomics i. Data Wrangling and Analytics with Tidyverse 1. Stop at Mutate. This will be covered next week. i. YouTube: @RProgramming101: 10 data filtering tips using R programming j. Advanced: duplicates and sampling & Indexing data manipulations i. Course Video ii. Course Video iii. Course Reading: FIND/WRITE: Lesson 7 2. Replicable Work a. Course Video: Formatting an Quarto (date, code chunks) b. Course Video: Write memos in Quarto c. Course Video: Render Quarto to HTML d. Reading: Quarto cheatsheet e. YouTube: @RProgramming101: Quarto crash course f. YouTube: @RProgramming101: Quarto YAML g. YouTube: @RProgramming101: Quarto Code Chunks h. YouTube: @RProgramming101: Quarto - replacing R Markdown
Activity or Assessment <i>Part 1</i>	1. Advanced: <u>Activity</u>: sample() and ifelse() in "Lesson 6 Point Boost" 2. Advanced: <u>Activity</u>: Build your own dataset in "Lesson 6 Point Boost" 3. Core: <u>Assignment</u>: Manipulations in "Lesson 7"
Activity or Assessment <i>Part 2</i>	1. Core: <u>Assignment</u>: dplyr Intro in "Lesson 8" 2. Advanced: <u>Quiz</u>: Extra Practice Base R + duplicates and sampling
Signature Assignment	1. Advanced: Explore selected dataset Also see the active Canvas course page "Explore to Plan." 2. Advanced: Indexing data manipulations. Also see active Canvas course page "Base R Data Transformations."
Module 6	
Title	R: Dplyr Verbs; LS: Openmindedness; AI techniques and applications
CLOs, MLOs	R • Core ○ Use dplyr verbs (mutate, case_when, summarize, group_by) ○ Assign factor class with levels to columns • Advanced ○ Compare dplyr with base R methods ○ Manually calculate and explain basic descriptive statistics ○ Build a pipeline ○ Create a GitHub repository linked to R Studio LS • Advanced ○ Open-mindedness Skill Highlight ○ Review course content AI • Advanced

	- Whether or not to use a specific AI system to achieve a justified aim. - Use vibe coding techniques
Reading / Video / Content	Part 1 - dplyr part 2 - Course Lesson: "Lesson 9" - Course Video: mutate() - Course Video: case_when() - Course Video: mutate() and case_when() - Course Video: factor() - Reading: Data Science A First Introduction 3.7 Using mutate to modify or add columns 3.10 Apply functions across many columns with mutate and across 3.11 Apply functions across columns within one row with rowwise and mutate - Reading: Data Analysis and Visualization in R for Ecologists Making new columns with mutate() - Reading: R for Social Scientists: Introducing dplyr and tidy (dplyr section) - Reading: Data Analysis and Visualization with R for Genomics Data Wrangling and Analytics with Tidyverse - Start at Mutate and finish page. - YouTube: @RProgramming101: Select Filter Mutate - Replicable Work - Course Reading: FIND/WRITE Open-mindedness learning skill - Course Reading: FIND/WRITE AI techniques and applications - Course Video: AI techniques and applications Part 2 - Dplyr part 3 - Course Lesson: "Lesson 10" - Course Video: summarize() - Course Video: group_by() - Course Video: multiple variables - Course Video: multiple statistics - Reading: Data Science A First Introduction 3.9 Aggregate data with summarize and map - Reading: Data Analysis and Visualization in R for Ecologists The split-apply-combine approach - Advanced: Course Lesson: "Lesson 10" Descriptive Statistics - Course Video: Descriptive Statistics - Advanced: Course Lesson: "Lesson 11" - Course Video: Building a pipeline - Reading: Data Science A First Introduction 3.8 Combining functions using the pip operator - Replicable Work - How to create a GitHub Repo: README - Course Video: How to manage GitHub (Windows) - Course Video: How to manage GitHub (Mac)???
Activity or Assessment Part 1	- Core: Assignment: dplyr 2: in "Lesson 9" - Advanced: AI techniques and applications activity - Advanced: Quiz: Extra practice
Activity or Assessment	Core: Assignment: dplyr 2: in "Lesson 10"

Part 2	2. Advanced: <u>Activity</u> : build a pipeline in “ Lesson 11 ” with Descriptive Statistics 3. Advanced: GitHub <u>Activity</u>
Signature Assignment	1. Advanced: FIND/WRITE : Compare dplyr with base R Indexing methods
Module 7	
Title	R: Data Storytelling LS: Thoroughness; AI: AI techniques and applications
CLOs, MLOs	R - Core - Use ggplot to create figures (scatter, histogram, boxplot, density) - Simulate probabilities - Advanced - Design tables LS - Core - Format Document - Advanced - Thoroughness Skill Highlight AI - Advanced - Use vibe coding techniques
Reading / Video / Content	Part 1 - ggplot2 basics - Course Lesson: “Lesson 13” - Course Video: data, aes, gemo - Course Video: scatterplot - Course Video: histogram - Course Video: boxplot - Course Video: density plot - Reading: Data Science A First Introduction 1.10 Exploring data with visualizations - Reading: R for Social Scientists: 5 Data Visualisation with ggplot2 - Reading: Data Analysis and Visualization in R for Ecologists Data visualization with ggplot2 - YouTube: @RProgramming101: Data visualization with ggplot ggplot to create bar charts for 2 categorical variables Bar charts and histograms ggplot playlist - Replicable Work - Course Reading: FIND/WRITE Thoroughness Learning Skill - Course Reading: FIND/WRITE AI techniques and applications - Course Video: AI techniques and applications FIND/WRITE : Vibe coding Part 2 - Tidyr - Reading: Data Analysis and Visualization in R for Ecologists Reshaping data with tidyr (tidyr section) - Reading: R for Social Scientists: Introducing dplyr and tidyr - Simulation

	a. Course Lesson: " Lesson 12 " b. Course Video: sample, rnorm c. Course Video: replicate d. Course Lesson: Fire
Activity or Assessment <i>Part 1</i>	1. Core: Assignment: ggplot2 in " Lesson 13 " 2. Advanced: Vibe-coding
Activity or Assessment <i>Part 2</i>	3. Core: Assignment: in " Lesson 12 " 4. Advanced: Fire Assignment
Signature Assignment	Advanced: Figures and tables

Grades

Categorize all **graded work** into assignments, points per submission, number of submissions, and percentage of total grade. You may want to use the [Template Grade Distribution Plan](#) or [Template Grade Distribution Plan with Grading Time](#). For help on deciding, review [Determining Grade Values or Percentages](#). For a deeper understanding of grade distribution, you may choose to [Prepare a Grade Distribution Plan](#).

Final grade levels will be A, B, C, D and F (no plus or minus). To receive an A in the course, the student must complete the A bundle. To receive a B, the student must complete the B bundle. To receive a C, the student must complete all Core activities and assessments.

		2 Core or Advanced Sig	2 Core or Advanced Sig
	Core Sig Ass.	50% Advanced	80% Advanced
Core Sig Ass.	Core	Core	Core
75% Core			
D Bundle	C Bundle	B Bundle	A Bundle

Students will have **five** tokens at the beginning of the semester to use for a revision or an assignment extension.